

3.5IV. DRAFT QUESTIONNAIRE

A. GENERAL INFORMATION ABOUT THE RESPONDENT:

- Country*: **Russian Federation**
- Organization(s) or entity(ies) responsible for the preparation of the report*:

Name: **Ministry of Science and Higher Education of the Russian Federation**

Website: <https://minobrnauki.gov.ru/>

Email address*:

Phone number*:

Please describe the role/mandate of your organization:

The Ministry of Science and Higher Education of the Russian Federation is a federal executive body that exercises functions on the development and implementation of state policy and legal regulation in the sphere of higher education and relevant additional professional education, scientific, scientific-technical and innovation activities, nanotechnology, the development of federal centers of science and high technology, state research centers and science cities, intellectual property (except for legal regulation of issues related to control, supervision and provision of public services in the sphere of legal protection of inventions, utility models, industrial designs, programs for electronic computers, databases and topographies of integrated circuits, trademarks, service marks, geographical indications, appellations of origin of goods), in the sphere of social support and social protection of students, youth policy in relation to young people studying and (or) working in organizations implementing educational activities under higher education educational programs and scientific organizations, as well as functions on the provision of public services and management of state property in the sphere of higher education and relevant additional professional education, scientific, scientific-technical and innovative activities, including the activities of federal centers of science and high technology, state scientific centers, unique scientific stands and installations, federal centers for collective use, leading scientific schools, a new generation national research computer network and information support for scientific, scientific-technical and innovative activities.

- Officially designated contact person(s) who completed this survey:

Full name*:

Position*:

Email address*:

- Full Name(s) of designated official(s) certifying the report:
- Brief description of the consultation process established for the preparation of the report:
Provided information was gathered from the sources of the Ministry of Science and Higher Education of the Russian Federation used in its activity.
- Other organization(s) or entity(ies) (including non-governmental) consulted for completing this survey: **NO**

Name of the organization*:

website:

Email address*:

Sector: ☐ Public ☐ Private ☐ Civil Society ☐ Other

B. Questionnaire for Member States' reporting on the implementation of the 2021 Recommendation

1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE, ASSOCIATED BENEFITS, AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

1.1 Has the 2021 Recommendation been promoted and/or shared with appropriate ministries and

institutions as well as affiliated organizations in your country?

YES/ NO

If yes, please indicate which ministries and/or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

The Ministry of Foreign Affairs of the Russian Federation, the Ministry of Science and Higher Education of the Russian Federation, the Russian Academy of Sciences.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

YES/ NO

If yes, which language(s)?

Russian

1.3 Have there been awareness raising activities on open science, including all its key elements, associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref.: (1) 16.f, 16.q, 16.h)

YES/ **NO**

If yes, please provide details on the activities organized or foreseen and links as relevant.

1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science, in publicly funded research? (ref.: (1) 16.a, 16.b, 16.c, 16.d, 16.e, 16h)

YES/ **NO**

If yes, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so?

DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

2.1 Does your country have a national policy, strategy or plan of action on science, technology and innovation (STI)?

YES/ NO

If yes,

- Please attach/upload a machine-readable PDF file of the policy and provide the following information:

Title of the policy **the Scientific and Technological Development Strategy of the Russian Federation**

the authority in charge of development of the policy **the President of the Russian Federation**
entities involved and the process of development

The implementation of state policy in the field of scientific and technological development is ensured by the coordinated actions of federal government bodies, government bodies of the constituent entities of the Russian Federation, local government bodies, the Russian Academy of Sciences, funds for the support of

scientific, scientific-technical and innovative activities, institutes for innovative development, scientific and educational organizations, industrial enterprises and other organizations directly carrying out scientific, scientific-technical and innovative activities and using the results of such activities, in accordance with their powers.

year of adoption **2016**

year of last update **2024**

key areas/sections of the policy

a) to form an effective system of interaction between science, technology and production, ensuring increased receptivity of the economy and society to new technologies, creating conditions for the development of knowledge-intensive entrepreneurship;

b) to create infrastructure and conditions for conducting scientific research and development, introducing knowledge-intensive technologies that meet modern principles of organizing scientific, scientific-technical and innovative activities, based on the best Russian and world practices;

c) to create opportunities for identifying and nurturing talented youth, building a successful career in science, technology and technological entrepreneurship, ensuring the preservation and development of the intellectual potential of science, increasing the prestige of the profession of scientist and engineer;

d) to form an effective system of management in the field of science, technology and production and investment in this area, ensuring a single scientific and technological space focused on solving state problems and meeting the needs of the economy and society;

d) to promote the formation of a model of international scientific and technical cooperation and international integration in the field of scientific research and development.

and links to the webpage if available <http://en.kremlin.ru/acts/news/73579>
<http://static.kremlin.ru/media/events/files/ru/HHNAzTI1guvX9Y00yaFA4KkMWPYcWS8.pdf>

- does it include any of the following open science elements:
 - open access to scientific knowledge
 - open science infrastructure
 - open engagement of societal actors
 - open dialogue with other knowledge systems.

Yes

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

2.2 In your country, is there a policy, or a set of policies and/or legal framework(s) at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

YES / **PARTLY** / NO

If yes or partly, please attach/upload machine-readable PDF file(s) of the policy(ies) and provide the following information for each:

Title of the policy or legal instrument **Concept of International Scientific and Technical Cooperation of the Russian Federation**

the authority in charge of development of the policy **The Presidential Council for Science and Education**

entities involved and the process of development

The Federal Assembly of the Russian Federation shall ensure the development of the regulatory framework for the International Scientific and Technical Cooperation.

The development of strategic, planning and program documents to ensure the implementation of the Concept is assigned to the Ministry of Science and Higher Education of the Russian Federation.

At the international, national, regional and municipal levels, individual functional areas and tasks related to the implementation of the International Scientific and Technical Cooperation, are ensured by:

- Federal Agency for the Commonwealth of Independent States, Compatriots Living Abroad, and International Humanitarian Cooperation (Rossotrudnichestvo), including its offices abroad;
- The Ministry of Science and Higher Education of the Russian Federation;
- The Ministry of Economic Development of the Russian Federation;
- The Ministry of Industry and Trade of the Russian Federation;
- the Russian Academy of Sciences;
- other federal executive bodies and state corporations;
- Russian development institutes and representative offices of companies with state participation abroad;
- leading organizations carrying out research and development, federal and national research universities and other leading universities.

At the regional level, the implementation of the provisions of the Concept is the prerogative of regional authorities responsible for scientific and technical/educational activities.

year of adoption 2019

year of last update It is planned to update the Concept in 2025

key areas/sections of the policy

- development of a system for training domestic and attracting the best world personnel, development of competencies in the field of scientific research, development and innovation, as well as international scientific and technical cooperation and science diplomacy;
- formation in Russia of a comfortable environment for scientific research, development and innovation within the framework of International Scientific and Technical Cooperation, including through the improvement of institutions, norms, support mechanisms and forms of organization of scientific and technical activities and International Scientific and Technical Cooperation;
- development of modern research and technological (including digital) infrastructure;
- creation of new forms of international scientific and technical cooperation, development of science diplomacy;
- ensuring within the framework of International Scientific and Technical Cooperation effective transfer and commercialization of the results of scientific and technical activities in the Russian and global economy;
- formation of international regimes and standards, institutions and processes that ensure the effective implementation of the International Scientific and Technical Cooperation of the Russian Federation.

elements of open science that are addressed

The international scientific and technical cooperation of the Russian Federation is carried out on the principle of openness.

Openness implies - with reasonable measures to strengthen the protection of intellectual property and other results of intellectual activity:

- freedom of scientific and technological communication;
- free, equal and non-discriminatory access to scientific information, research results, scientific infrastructure, competencies, as well as individual technological developments that are important for solving humanitarian and other universal problems.

and links to the webpage(s) if available

https://minobrnauki.gov.ru/upload/2021/07/kontsepsiya_MNTS_Rossiyskoy_Federatsii.pdf

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policies.

2.3 In your country, are there specific policy instruments that aim to promote open science?

YES/ UNDER DEVELOPMENT/ **NO**

If yes or under development, please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17h)

YES / UNDER DEVELOPMENT / **NO**

If yes or under development, please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at institutional level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref: (ii) 17.d, 17.e, 17.g)

YES /UNDER DEVELOPMENT/ **NO**

If yes, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and elements of open science included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors).

2.6 In your country, are there specific funding mechanisms for open science at the national level? (ref.: (ii) 17.i, (iii) 18.i, (iv) 19.c)

YES / UNDER DEVELOPMENT / **NO**

If yes or under development, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such funding mechanisms.

2.7 In your country, have open science practices been integrated in existing research funding mechanisms?

YES/ **NO**

If yes, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and

indicate if there are plans to do so.

2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: (ii) 17. i)

YES/ **NO**

If yes, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy? (ref.: (ii) 17.i, 23.c)

YES / UNDER DEVELOPMENT / **NO**

If yes or under development, please provide details and links as relevant.

3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)

0.96%

2024

2023

<https://issek.hse.ru/mirror/pubs/share/993677988.pdf>

3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.b)

87.9%

2024

2023

<https://issek.hse.ru/mirror/pubs/share/995751983.pdf>

3.3 Are there national research and education networks (NRENs) active in the country? (ref.: (iii) 18.c)

YES/ NO

If yes, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

- Please provide the following information:

title of networks or legal instrument **A new generation National Research Computer Network (NRCN)**

the authority in charge of development of the policy **Ministry of Science and Higher Education of the Russian Federation**

key areas/sections of the network

- providing opportunities for scientific and educational organizations of higher education to conduct research and development in priority areas of scientific and technological development, implementation of modern ICT in the realization of educational programs of higher education, participation in major Russian and international scientific and educational projects;

- participation in the formation and functioning of a unified scientific and educational information space of the Russian Federation in order to effectively use the advanced infrastructure of research, development and innovation activities being created in the country, including to ensure global accessibility and distributed operation with big data;
- ensuring departmental network connectivity and provision of a complex of specialized services for scientific and educational spheres, implementation of development of interfaces for managing own network resources for integration with digital platforms of science and education including the use of federated authentication technologies.

year of adoption 2019

entities involved and management organizations

The management of the NRCN is assigned to the Ministry of Science and Higher Education of the Russian Federation. The Council for Digital Development and Information Technologies under the Ministry of Science and Higher Education of the Russian Federation stands as the collegial management body of the NRCN.

Users of NRCN services include:

- scientific and educational organizations, organizations with mega science class facilities;
- member states of world-class research and education centers, Competence Center of National Technology Initiative, Technology Transfer Centers;
- the Ministry of Science and Higher Education of the Russian Federation;
- the Russian Academy of Sciences, branch academies of science and other organizations subordinated to the Ministry of Science and Higher Education of the Russian Federation;
- federal executive bodies involved in research and development activities;
- organizations of the real sector of the economy engaged in scientific and technological activities and others.

and links to the webpage if available https://niks.su/docs/management_procedure_2021.pdf

Title of networks or legal instrument State Information System “Modern Digital Educational Environment” (MDEE)

the authority in charge of development of the policy Ministry of Science and Higher Education of the Russian Federation

key areas/sections of the network

- formation and maintenance of a register of online courses implemented by a variety of educational organizations;
- integration with educational platforms and information systems of educational organization, as well as organization of centralized accounting of training achievements in online courses;
- accounting of educational programs of higher education and additional professional programs as well as academic results.

year of adoption 2020

entities involved and management organizations

The Ministry of Science and Higher Education of the Russian Federation establishes the procedure for creating and maintaining a register of online courses.

The participants of the system are the Ministry of Science and Higher Education of the Russian Federation and users of the system (individuals and legal entities):

- educational organizations that are right holders of online courses and operators of educational platforms;
- experts who examine the online course and evaluate the quality of the online course;
- students;
- employers.

and links to the webpage if available <http://government.ru/docs/all/130930/>

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to establish them.

3.4 Are there national or institutional open science infrastructure in your country? (ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)

YES / UNDER DEVELOPMENT / NO

If yes or under development, please indicate which type(s) of infrastructure:

- federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage.
- community managed infrastructures, protocols and standards, for example those that support biodiversity and engagement with society.
- platforms for exchanges and co-creation of knowledge between scientists and society
- community-based monitoring and information systems to complement national, regional and global data and information systems.

For each infrastructure, please provide links as relevant and more information with regard to their compliance with 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community, governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective, Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

Type(s) of infrastructure **federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage**

Title of open science infrastructure or legal instrument **the Megascience class scientific facilities**

the authority in charge of development or implementation of the policy **Ministry of Science and Higher Education of the Russian Federation**

General information about open science infrastructure

Creation of an advanced infrastructure for research and development and innovation activities, including the creation and development of a network of unique mega sciences class scientific facilities is implemented in accordance Executive Order on National Goals and Strategic Objectives of the Russian Federation through to 2024.

In 2023, there were eight mega sciences class projects in Russia, some of them are:

- International center for neutron research based on high-flux research reactor PIK;
- Complex of superconducting rings for heavy ion colliding beams (NICA complex);
- Kurchatov specialized synchrotron radiation source "KISI-Kurchatov";
- Siberian ring photon source" (TsKP "SKIF");
- Synchrotron radiation source "SILA";
- Synchrotron radiation source "RIF", etc.

entities involved and the process of development **Ministry of Science and Higher Education of the Russian Federation, organizations with mega science class facilities and others**

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref: (iii) 18.e, 18.f)

YES / NO

If yes, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science, stage of research, accessibility to all the relevant users in the region or internationally).

Title of the infrastructure **the Joint Institute for Nuclear Research (JINR)**

Date of establishment **March 26, 1956**

Language **Russian and English**

Governance/community agreements

JINR is a legal entity and possesses on the territory of the member states the legal capacity necessary for the realization of its functions.

In accordance with its status as an intergovernmental organization, JINR has international legal personality, in particular, it has the right to conclude international agreements and to have observers at other international organizations, the activities of which correspond to the statutory objectives of JINR.

Scope:

- *research areas*
Particle physics, nuclear physics and condensed matter physics
- *elements of open science*
 - **JINR Laboratories;**
 - **scientific research base (experimental facilities for physics, powerful high-productive computing environment);**
 - **international cooperation;**
 - **educational activities;**
 - **open access publications.**
- *stage(s) of research*
 - **conducting theoretical and experimental research in the accepted scientific directions of its activity;**
 - **exchange of experience in conducting scientific research, as well as information obtained as a result of this research, by publishing scientific papers, holding conferences, symposiums, organizing internships;**
 - **establishing and maintaining contacts with other national and international scientific institutions and organizations in order to coordinate activities and organize mutually beneficial cooperation.**
- *accessibility to all the relevant users in the region or internationally*
Participation in JINR activities is implemented in diverse forms: on the basis of membership in JINR, as well as bilateral and multilateral agreements on participation in the implementation of individual scientific programs. On the basis of agreements concluded with JINR, international and national research organizations and institutions, as well as individual scientists from non-member states may participate in JINR's scientific projects. In order to exchange information, establish cooperation and coordinate activities, international organizations might be granted observer status with the JINR.

and links to the webpage if available **<https://www.jinr.ru/about-en/>**

If no, are the existing open science infrastructures at regional and international levels accessible to all the relevant open science actors from your country?

YES / NO

3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: (iii) 18.g)

YES / **NO**

If yes, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science, stage of research, accessibility to all the relevant users in the region or internationally).

4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: (iv) 19.a, 19.c)

YES / NO

If yes, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

4.2 Have capacity building programs for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

YES / NO

If yes, please provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

4.3 Has a framework of open science competences been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: (iv) 19.b)

YES/ NO

If yes, please provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

The development and approval of educational programs are within the competence of the educational organization. Organizations develop educational programs in accordance with federal state educational standards. Thus, issues of the content of educational programs and the inclusion of competencies in them are attributed to the powers of the educational organization.

4.4 Are open educational resources as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: (iv) 19.d)

YES/ PARTLY/ NO

4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researches of different disciplines, decision-makers and the public at large? (ref.: (iv) 19.e)

YES / NO

If yes, please provide more information and links as relevant.

Title of initiatives or legal instrument **the Decade of Science and Technology in the Russian Federation**

the authority in charge of development of the policy **the President of the Russian Federation**

key areas/sections of the policy

- **attracting talented young people to the area of research and development;**
- **promoting the involvement of researchers and developers in solving the most significant**

social and national development issues;

- increasing accessibility of information on the achievements and prospects of Russian science for citizens of the Russian Federation.

year of adoption 2022

entities involved and the process of development

The Government of the Russian Federation approves the plan for the Decade of Science and Technology in the Russian Federation.

Federal executive bodies ensure the implementation of the plan's activities and develop departmental plans of the main activities for the Decade of Science and Technology in the Russian Federation.

Regional government bodies are involved in the implementation of the Decade of Science and Technology in the Russian Federation and develop regional plans of the main activities for the implementation of the Decade of Science and Technology in the Russian Federation. The plan for the Decade of Science and Technology in the Russian Federation contains information on the names of initiatives (projects and activities), information on indicators and executors of initiatives.

The promising initiatives, projects and events of the Decade Plan include: the initiatives "Science is Nearby", "Science of Winning", "Scientific Volunteering", "Third Semester", "Working with Experience and Designing the Future", "Science and Business", "Popular Science Tourism", "Science as Art", "Solutions and Services for the Professional Community", the "Scientific Children's Playgrounds" project, thematic initiatives of the Decade of Science and Technology on the priorities of the Strategy for Scientific and Technological Development of the Russian Federation, as well as the Congress of Young Scientists.

In 2024, over 12,000 federal and regional events were held as part of the Decade. More than 43 million people took part in the projects and events.

As part of the implementation of the initiatives "Science Nearby", "Science of Winning", "Scientific Volunteering" and "Third Semester" over 200 lectures, open lessons and excursions to scientific laboratories were held. More than 2 million schoolchildren took part in the Olympiads. Over 100 scientific projects for young scientists were also implemented. The initiatives "Working with experience and designing the future" and "Science and business" made it possible to implement over 24 thousand start-up projects and demonstrate promising developments of young scientists.

As part of the grand competition for the production of national cinema, a separate track was launched for the first time for films dedicated to modern Russian science and outstanding scientists of the past (the "Science as Art" initiative).

Also, 76 popular science tourist routes and 47 scientific playgrounds have been opened in the Russian Federation.

The IV Congress of Young Scientists brought together over 7,000 participants.

In 2025, within the framework of the Decade, work will continue on a number of projects, including: support for science fiction, development of a literature fund and creation of a special thematic library, integration of science into art, and development of the artistic image of a scientist in libraries, museums, theaters.

The Ministry of Science and Higher Education of the Russian Federation has developed and is implementing support measures aimed at popularizing science and scientific knowledge. For the first time in 2023, a competition was organized to support authors covering issues of science and technology. In March 2024, following the results of the grant competition for authors, 19 agreements were concluded.

For the first time in 2023, a competition was organized to support organizers of popular science events for a wide audience. In March 2024, following the results of the grant competition for event organizers, 18 agreements were concluded.

and links to the webpage if available <https://xn--80aa3ak5a.xn--p1ai/>

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref: (v) 20.b, 20c, 20.i)

YES / **NO**

If yes, please provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref: (v) 20.a, 20.c, 20.d)

YES / **NO**

If yes, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20.h)

YES / **NO**

If yes, for each initiative, please provide more information, including on the stakeholders that are involved or lead the initiative, and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

5.4 Have there been any unintended negative consequences of open science practices reported in your country? (ref.: (v) 20)

YES/ **NO**

If yes, please provide more information and specify if there is a strategy or plan to preventing and mitigating such consequences.

6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: (vi) 12)

YES, at the national level.

YES, at the institutional level.

YES , at both national and institutional levels.

NO

If yes, please indicate which of the recommended actions they address and provide more information and links as relevant:

- promotion of pre-prints
- exploration of open peer-review practices
- valuing and sharing of negative scientific results and associated data
- participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects
- development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas
- promoting open innovation practices.
- Others (with a box to add text)

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 recommendation on Open Science? (ref.: (vii) 22.a)

7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration in open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e, 22.f)

YES / **NO**

If yes, please provide links as relevant and more information.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: <https://unesdoc.unesco.org/ark:/48223/pf0000379949/PDF/379949eng.pdf.multi.nameddest=>)

YES / **NO**

If yes, please provide links as relevant and more information.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

8. GENERAL CONSIDERATION

8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

YES/ NO

If yes, please provide more information.